Format

- Case reports
- visual fields
- glaucoma
- 2 or 3 anterior eye cases

Anterior Eye

- **Usually…CLMK, CLPU, HSV epithelial, HZO, HSK, hyphaema, infiltrative keratitis, AAU**
    - No difficult cases- won't be SLK, Thygeson's, weird viral
- **AAU**
    - **Stellate KP**
- **Herpes Simplex Epithelial and Stromal**
- **Herpes Zoster**- Henessey
- **MK/Bacterial Keratitis**
- **EKC**- Henessey
- Bacterial conjunctivitis
- Chlamydial conjunctivitis
- Blepharoconjunctivitis
    - Secondary to alpha-agonists (Henessy)
- Marginal Keratitis
- Phlyctenulosis
- VLK
- Hyhaema

Glaucoma

- Possner Schlossman
- AAC
- Steroid induced IOP rise
- PXG and VF differences
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Combos

- Hyphaema → blunt trauma → angle recession glaucoma
    - Mx and tx
    - List all consequences from anterior to posterior
- Marginal keratitis and NTG
- Ocular allergy and NTG
- CLPC and PXF

- PXF Glaucoma
- NTG (with a temporal wedge)
- Blunt trauma – hyphaema + angle recession
- Acute angle closure glaucoma ACG
- Visual pathway visual field defects & NTG, 6 visual fields
- PDS, PDG + NTG

- Fuch's/Possner Schlossman and PXG
- POAG
- NTG (care sectoral ONH pallor ddx neurological/compressive problems)
- Inflammatory OAG

MK 2017 questions (5-7 cases per person)

- Neuro

- Diabetes
- Palsies
- Glaucoma
- Interpreting OCT related to diabetes
- Uveitis
- Conjunctivitis
- Kinetic perimetry
- Difference between a bacteria and human cell
- Why does herpes cause a lid ptosis
- Blood supply
- How do you culture MK
- Vortex keratopathy
- AAU
- Contact dermatitis and allergic conjunctivitis
- HZO
- Visual pathway lesion → what defect
- =Internuclear ophthalmoplegia
- = Iridodialysis
- Pseudopapilloedema

Tips

- Say as little as possible to answer a question- they will clarify if they need more
- Reference all GLAUCOMA trials (EMGT less importance) and maybe SCUTS and SUN

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Anterior Eye

- Treatment
    - Steroids: Never ever put steroids as first line of therapy and ESPECIALLY DON'T USE FOR HERPES AND CL-MK!
        - Only use when wound has healed and when there is no more infection
        - Before giving steroids….need to ask 3 questions!
            - Ask HSV active stage? Yes then don't give!
            - Ask glaucoma? More likely a steroid responder!
            - Fhx glaucoma? First-order relative then 50-90% increased risk of being a high steroid responder
        - Steroid responders will respond faster with IOP increases than other people, and other people will also get higher IOPs but to a lesser extent
    - Loading Dose: Always need a loading dose for MK infections – every 15 mins for first hour then every hour for first 6 hours (follow Will's Eye Manual – double check this)
    - Use generic names for all drugs, state concentration, 1 drop or how many drops, dosage, review period
        - E.g. every 1 hour then will follow up the next day
    - Pain relief
        - Ibuprofen 200mg, Atropine 1% 1-2x/day or Cyclo
    - Note that we can't prescribe for episcleritis and scleritis because need 400mg and 800mg dosages so need GP. Medicolegally need to caution against patient just taking 2 of the tablets to get around this!
    - Always know mechanism of action of the drugs!
    - Fluoroquinolones: Side effects include white precipitates on ulcer (17%), athropathy (weakness of tendons behind ankles)
        - Fluoroquinolones are DNA gyrase inhibitors. DNA gyrase is involved in super coiling of bacterial DNA, allowing the bacteria to replicate
    - Note there can be a compounding effect with having an oral and topical drug at the same time à NSAIDs & beta blockers most common/worrying class for this
- Corneal Ulcer: Describe size, circular vs dendritic, well-defined or not, location
    - Cyclo doesn't usually help with the 'pain' because dilates and causes photosensitivity
    - Case may start off not central etc and then after you manage that they may change it up and say what if the ulcer was now central (more severe)? Then you need to send to hospital for foritified cephalosporin and aminoglycoside
    - Another e.g. is a herpes dendritic ulcer becomes a stromal ulcer
    - Corneal epithelial cells will not re-epithelialise if the nerve does not produce growth factor. And the nerve may not produce growth factor if there is neurotrophic keratitis! So deinnervation of a cornea will lead to corneal perforation. Note that corneal deinnervation can be mild or severe so level of severity will vary accordingly
- Blunt trauma
    - Know front of eye to back of eye the effects of blunt trauma!
- Episcleritis is usually young females!!! So don't just throw that out as number 1 ddx if doesn't fit the age/gender

Posterior Eye

- **Usually… glaucoma!!! No-one has ever had AMD etc before**
- Describe the disc
    - No drance haems
    - Which rim looks thinner, superior or inferior
    - Arteriole attenuation/calibre associated with glaucoma
    - RNFL reflectivity
        - Compare superior and inferior reflectivity
    - True bayonetting is when there is no rim left!
    - Pallor? Be wary that a disc can look pale due to glial tissue filling the space (e.g. optic atrophy). If no rim left then can also result in a pale looking disc but not actually pallor
- Visual Fields
    - Reliability: Arbitrary criteria, no right answer, read one study, pick that criteria and stick with it and quote it
        - E.g. <33% FP and FN, and <20% fixation losses (Phu et al 2017..??)
    - Mention 'inattention' not malingering
    - Blind spot monitor: Discs with a big PPA or large discs will usually not have fixation losses
    - Gaze tracker: Uses infrared camera, big pupil can make gaze tracker go crazy because it uses the Purkinje reflections. The lines going up are the eye movement, and down is blinking
    - False negatives not used as a reliability index in ANY glaucoma study à median false negative rate in glaucomas is 40% but in normal should be less than 6% (M.K. quoted Angus Hale etc as 1% for normals and 6% for glaucoma though). False negs will be really high in patient with severe visual loss
        - High false negatives do not make fields unreliable!!!
        - False negatives, need to confirm whether it is non-physiological visual field loss (i.e. inattention)
    - Mean Deviation <12
        - -3, -6, -12 à depends on staging system à Mill's Criteria, AAO, pick one!
    - PSD: Increased deepening of defects, can't get lower than 0, so PSD gets lower in end stage glaucoma
        - Useful in early stages but then becomes useless as it doesn't get worse with sever vision loss, whereas MD does J
    - GHT: Probability analysis, uses 5 areas, compares probability values (not the threshold value numbers!!!), so if above and below the horizontal line are marked as p<0.5% then it may still say that the GHT is normal! But… if the threshold value is really really bad then there is a clause for this in the machine to say that GHT is abnormal
        - Know the 5 zones! Can be asked to trace them out
    - VFI: Ignore this. May have to describe what it is but there is no pass or fail criteria with is!
- DDx's
    - Optic neuritis always is a ddx, as well as trauma and compressive optic neuropathy
    - NTG, POAG, ACG, NA-AION, past compressive optic neuropathy, trauma, optic neuritis, secondary glaucomas (PXF, PDS, angle recession, Fuch's, inflammatory, neovascular)
    - Not usually uveitic or inflammatory glaucoma cases given, but will do PXF
    - Two conditions with typically deep cupping are NAAION, AAION and glaucoma
    - Congenital glaucoma would see a big eye (buphthalmos)
- Glaucoma
    - CCTs

- Phasing IOP: Varies by 2-4mmHg diurnally, iCare 2.1-7.6 glaus and 2-6 normals

- Treatment
  - Initiate treatment then referral letter must be within 4 months!
  - Always initiate in both eyes if both eyes have glaucoma
    - Say you would do a monocular trial though if you get Isabelle as examiner
    - Can do monocular trial if want to try out side effects or if the other eye really don't have glaucoma
    - If the other eye is completely blind then there is no need to treat unless having pain from neovascular glaucoma for example
  - Don't ever do combo drugs first, follow the NHMRC guidelines
  - Know the side effect profiles and contraindications
    - Beta blockers in particular
    - Obscure side effects like headaches don't really need to know
  - Note that there can be compounding effect with having an oral drug and topical drug at the same time à NSAIDs and beta blockers are the most worrying classes for this compound effect
- Pregnancy!!!
  - Pharma Phu et al
- They might then change up the glaucoma case
  - E.g. from big VFD to clear VFD (but same structural loss) how would you manage differently?
  - E.g. Normal IOP to high IOP how would you manage differently?
- IOP
  - Patients can start to feel it is high from 30+
  - If it is 40-50mmHg then can be a few months before it has noticeable damage (I think?)
  - If it is IOP 60+ then can have damage within 48-72 hours
- ACG Attack
  - Call Ophthal first then GP would be next line if unavailable, then hospital
  - Acute attacks there would be VFDs but not blind
  - Chronic attacks more likely larger VFDs
- 3 reasons why pressure is elevated in an eye
  - Outflow problems due to junk build up in drainage pathway
  - Increased aqueous production
  - Episcleral venous pressure (part of an outflow problem)

Visual Fields

- Left parietal lobe tumour à right lower quadrantanopia
- Left parietal lobe complete cut à right hemianopia
- Complete cut after chiasm on left side à Right hemianopia
- RAPD occurs if lesion is before the LGN before fibres loop
- Macula sparing = must be a cortical lesion
- Altitudinal defects 'altitudinal like' can be due to GCA (AION), glaucoma, occlusive events. Don't say adenomas for altitudinal though!!! Because an adenoma at the chiasm means bitemporal hemianopia, just that if its pituitary adenoma then visual field loss is stronger/starts from above (like a quadrantanopia)
- Acute papilloedema:
  - Enlarged blind spot
  - Chronic papilledema can cause arcuate loss
- ONH Drusen: Arcuate loss à calcified deposits so the tissue dies
- Glaucoma: arcuate loss and others
- Retinal arterial occlusive disease: Vascular arcades can cause arcuate loss

- Vertical midline – patient can't fake that and know where that line is since they are given random presentation!
- Describe like this: left absolute defect, right temporal hemifield and superior altitudinal (if it looks like a ¾ pie shape)
- Pituitary adenoma, visual field loss from above
- Cranopharyngioma, visual field loss from below
- Enlargement of blind spot would be all around the disc, not just some points next to it
- If a -3DS lens helps to reduce visual field defects this is because the patient has a myopic staphyloma or coloboma (developmental anomaly of incomplete closure of the cleft). If there was a real neurological deficit then VFD would not disappear with a -3DS lens!

Retinopathy vs Optic Neuropathy?

- Both can have decreased VA
- Usually photostress recovery time is 30 seconds, but with retinopathies will usually have a longer recovery time of >60 seconds (though can happen with both)

Random other things to know

- Diabetic à CN3 nerve palsy due to ischaemic attack with pupil involvement
- Lids = superior division of CN3 innervates LPS; Sympathetic nerve innervates Muller muscle. CN7 innervates orbicularis oculi
- Marcus Gunn syndrome = eye dilates in dim light then shine light and doesn't construct
- RAPD
- Advanced optic nerve problems
- Compressive à pupil fibres are located on the surface of the nerve so in an ischaemic attack they are still being given blood by CSF whereas a compressive lesion would squash the outside of the nerve so pupil would be affected
- Port-wine stain is a hamartoma which is a usually benign neoplasm of the tissue it originated from
- HZO: Lid vesicles on top eyelid. The most painful condition. Opthalmic division of eye involved. Acyclovir 800mg 4-5x, aim is to minimize PHN and its duration by treating HZO within 48 hours
- Infraorbital nerve travels along infraorbital notch
- Blow out fractures of orbits take 3 months to heal, lose feeling to nostril
- Hamartoma can affect episcleral venous pressure and cause higher pressures
- Sturge-Weber syndrome à can have anterior vascular anomalies in frontal lobe and can die
- Every patient that has had corneal injury will have cells in the anterior chamber around 12 hours after
- Hollenhorst plaque, ddx myelineation. This is an occlusive event
    - Relatively common retinal finding in the elderly, often visually asymptomatic and present with retinal emboli from plaque ulceration in internal carotid artery
- However, ocular emboli are rare! This is because the eye is fed by the internal carotid artery, first branch is the ophthalmic artery, and this comes off at a right angle so emboli wouldn't be able to go through there
- IOP decreases in uveitis due to CB shutdown, then increases again due to inflammation of TM (trabeculitis)

GO THROUGH ANTERIOR THERA QUESTIONS DOC

GENERAL RED FLAGS

1. NEVER give a topical steroid for microbial keratitis until there is marked improvement with the antibiotic or anti-viral meds.

2. Always taper a topical steroid if used long term. Also check IOP.

3. Never taper an antibiotic to less than 4 times a day because of antibiotic resistance.
4. Never give a non-selective beta blocker (Timolol) to persons with asthma, severe COPD or cardiac problems.
5. Never give PGA for persons with a history of uveitis, HSVK or CMO.

Other conjunctivitis
Please make sure to take careful history when checked for other conjunctivitis (e.g. sexual history).
Remember that conjunctival disease can occur in the presence of other ocular or system disease as well. E.g. conjunctival desiccation may be a concurrent sign in keratoconjunctivitis sicca, which may be associated with Sjogren's syndrome or other autommunie disease (Liu et al 2014 Curr Allergy Asthma Rep). Be sure that you can ask the appropriate questions to check for these (similar to review of systems performed in uveitis/scleritis). On that note, please also know the method for screening all 12 cranial nerves (important oral exam question).

General exam notes:
Differential diagnosis lists should begin with the most likely and end with the least likely. Focus on why it is NOT a certain disease/diagnosis, rather than spend a lot of time describing the disease itself (we know what it is). Remember it is critical to NOT be reliant on symptoms for diagnosis: look for signs.
Work-up should include all relevant work up for the immediate diagnosis and management of the case. Things like dilated fundus examination in a patient with uninvolved posterior eye or where posterior eye involvement is unlikely is superfluous and can be done at subsequent visits. Remember that ocular imaging can be useful in documentation of disease progression (e.g. Schiempflug imaging, anterior OCT, topography, photography, confocal). Look for specific signs in specific regions, e.g. pre-auricular lymphadenopathy. Check for other systemic signs and symptoms (review of systems, temperature, fever, joints etc).
Management plans should always include caveats. This means that you should plan for unexpected contingencies, e.g. "if the patient doesn't respond to treatment, I hypothesise that it could be xxx disease and therefore, I would perform xyz test again to rule out xxx disease…" In this vein, you MUST always have a review period of reasonable length; it is better to be slightly conservative if you are not confident on the diagnosis. Clinical course/prognosis should focus primarily on your diagnosis. Remember that any prescription of drugs, dietary modifications, lifestyle modifications etc should have some dosage included: you must always include drug concentration, dosage and duration (even if it is a preliminary, interim duration before you alter dosage). As a general rule of thumb: use generic names, not trade names. However, we argue that you should always tick "Brand substitution NOT permitted" (your mileage may vary).

Guidelines

- For each of the common conditions be clear about **management plan** including **treatment protocols (which drug, what dosage, frequency, for how long and when you want to review the patient, other issues including taper of steroid), ongoing management and advice to the patient**
- Know the **end points** for treatment
- Be prepared to **provide options** if the initial treatment is not successful
- Know the evidence for your treatment strategy – eg **OHTS, CNTGS, CITGS, AGIS, HEDS** etc.

- Understand the **risk factors for disease and risk factors for progression** particularly for glaucoma
- Be able to discuss **contraindications and local/systemic side effects** of particularly each of the **classes of glaucoma medications**. Know the **major drug interactions.**
- Contraindications and local/systemic side effects of other classes of S4 drugs – use in pregnancy, children etc, major drug interactions
- Know how the drugs work
- Know about the relative strength of different steroids and choice of formulation and route of administration.
- Don't get into arguments in the exam……if your opinion differs from that of the examiner, present the evidence base (hopefully not the lecture notes but the primary source) for your rationale
- Red flag topics would be fairly self evident but would include contraindications to glaucoma medications, treatment of herpetic disease, ocular emergencies and knowing when to refer
- Commonest errors – undertreating, activity of antibiotic agents, alternatives to fluoroquinolones

History
- Ask examiner questions

Investigation
- Always check VA, entrance tests

Diagnosis
- Group in order. E.g. cornea, conj
- Link signs and sx- photophobia could be due to compromised tear film, compromised cornea, compromised CB (uvea)
- Look at where to condition is, some may be a bystander rereaction

Differentials
- Make reasoning behind ruling out each differential
- DDx condition to help troubleshoot should the condition not improve with your treatment
- Always include big differentials e.g. HSV/MK

Aetiology/Pathophysiology
- What kind of organism is involved if infection
- What kind of chemical reaction involved in inflammation
- Tx is designed to target something along this path or cellular structure of organism

Management
- Name of drug
    - If giving oral drugs, specify "I will refer to GP and organise for this patient to obtain oral medication"
- Concentration of drug
    - If notes say 250-500mg decide on either 250mg or 500mg
- Dosage of drug
- Mechanism of Action
- Instruction for use
- Pregnancy class
    - Doxycycline has devastating effects
    - For males, preganancy irrelevant
- Contraindications
- Adverse Effects
- Patient education
    - E.g. chlamydial infection is a sexually transmitted infection- discuss sexual hx and educate patient about partners.. Get them tested

- If referring- what else can you do?
    - Symptomatic relief
        - Irritation- preservative free ocular lubricants
        - Photophobia- sunglasses

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2016

Combos

- Uveitis and steroid induced IOP rise
- Angle recession and AAU x2
- AAU and PXG
- Angle recession and MK (bacterial)
- NTG and MK (bacterial)
- Blepharoconjunctivitis secondary to alpha agonist and blunt trauma (hyphaema and angle recession)
- Herpes Zoster and acute angle closure
- Blunt trauma and hyphaema and angle closure and AAU
- Visual pathway, visual field defects + NTG, AAU
- PDS + NTG + MK
- Trauma + angle recession, Chlamydial conjunctivitis
- PXG + VF differentials, AAU
- PXG and EKC
- VF ddx, AACG, stellate KP AAU
- MK, marginal keratitis, phlyctenulosis, VLK, PDS, PDG, NTG (Andrew)
- Anterior uveitis, Fuchs/Possner and PXG
- Angle closure and MK
- Stellate KP Uveitis and POAG
- AAU + steroid response + PXG
- HZO + ACG, lots of fields, binasal hemianopia
- Visual pathway, PXG, AAU
- NTG (ddx neuro/compressive problems care sectoral ONH pallor), bacterial keratitis x2
- AAU, inflammatory OAG, possible Possner Schlossman, POAG
- AAC, 6VF, AAU (possible Possner) x2
- Blunt trauma, hyphaema, angle recession glaucoma, MK
- PXF, VF ddx, bacterial conjunctivitis x2
- Angle recession and chlamydial conjunctivitis

Number of drops in 1mL

· Depends on size of opening of dropper ranging from 30-50uL. Hence roughly 20 to 30 drops in one mL

Ocular Lubricants

- discuss / understand the factors you would weigh up and consider in making that decision (e.g. preserved versus non preserved, active ingredient that promotes wound healing or minimises shearing, desired viscosity, etc. and why).
- Don’t need to know all the ingredients of all the ocular lubricants by heart. You should know about the various possible active ingredients and the various components of formulations used in lubricants.